

## **Action Plan(by:- Lakshayaa Sundararajan)**

### **Project Overview**

In this project, we aim to use data science techniques to analyze YouTube trending videos. The project will include data collection, preprocessing of data, Exploratory Data Analysis (EDA), Natural Language Processing (NLP), time series forecasting, and visualization and documentation of the results. This action plan will be with a structured workflow of completing the project.

### **Data Acquisition**

YouTube Trending Video Stats was chosen from Kaggle because of the information it provides including video title, video channel name, video category, publish date, tags, views, likes, dislikes, comments, and other video metadata. The data was downloaded and loaded into Pandas in Python to be analyzed.

### **Data Preparation**

The data has been cleaned to enhance the quality of the data set before analysis. Records that were duplicated were deleted, missing values were addressed, and date columns were converted to the proper format. Data types were checked and cleaned data was ready for visualization and model building.

### **Exploratory Data Analysis**

The Exploratory Data Analysis (EDA) was used to gain an understanding of the nature of videos that go viral on YouTube. The analysis covered the study of the distribution of categories, top-performing channels, views, likes and dislikes, comments, publishing trends and relationships between various important numerical variables through charts and graphs.

### **Natural Language Processing**

The textual information found in video titles and tags were processed using Natural Language Processing techniques. The text preprocessing was done before doing sentiment analysis and keyword extraction with the TF-IDF. These methods helped to find trends and commonalities, and to grasp the character of trending content.

## **Time Series Forecasting**

Historical view data were sorted in chronological order producing a time series. The historical observations were used to predict the future viewing trends through Exponential Smoothing. Existing data was compared to the predicted values in order to gain insight into potential future trends in audience engagement.

## **Documentation and GitHub Repository**

All of the source code, visualisations, and observations were consolidated into a Jupyter Notebook after the analysis. The project files were placed in a GitHub repository with the README file that describes the project, dataset information, software requirements, and how to run the project.

## **Timeline**

Dataset Collection Day-1

Data Preparation Day-2

Exploratory Data Analysis-Day 3

NLP Analysis- Day 4

Time Series Forecasting- Day 5

Documentation and GitHub-Day 6

Final Review and Submission-Day 7

## **Expected Outcome**

The project aims to offer insights into the trends and patterns of audience engagement with videos on YouTube, as well as to predict future viewing trends and analyze text data. The final deliverables include a complete Jupyter Notebook, GitHub repository, research report, action plan, and presentation.

## **Conclusion**

This action plan outlines a systematic and clear way of completing this YouTube Trending Videos Analysis and Forecasting project. By adhering to these steps, the analysis can be structured, dependable, and generate useful insights from the existing data.